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RESEARCH PAPER

Phenomic data-driven biological prediction of maize through field-based high-throughput phenotyping integration with genomic data

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Abstract

High-throughput phenotyping (HTP) has expanded the dimensionality of data in plant research; however, HTP has resulted in few novel biological discoveries to date. Field-based HTP (FHTP), using small unoccupied aerial vehicles (UAVs) equipped with imaging sensors, can be deployed routinely to monitor segregating plant population interactions with the environment under biologically meaningful conditions. Here, flowering dates and plant height, important phenological fitness traits, were collected on 520 segregating maize recombinant inbred lines (RILs) in both irrigated and drought stress trials in 2018. Using UAV phenomic, single nucleotide polymorphism (SNP) genomic, as well as combined data, flowering times were predicted using several scenarios. Untested genotypes were predicted with 0.58, 0.59, and 0.41 prediction ability for anthesis, silking, and terminal plant height, respectively, using genomic data, but prediction ability increased to 0.77, 0.76, and 0.58 when phenomic and genomic data were used together. Using the phenomic data in a genome-wide association study, a heat-related candidate gene (*GRMZM2G083810*; *hsp18f*) was discovered using temporal reflectance phenotypes belonging to flowering times (both irrigated and drought) trials where heat stress also peaked. Thus, a relationship between plants and abiotic stresses belonging to a specific time of growth was revealed only through use of temporal phenomic data. Overall, this study showed that (i) it is possible to predict complex traits using high dimensional phenomic data between different environments, and (ii) temporal phenomic data can reveal a time-dependent association between genotypes and abiotic stresses, which can help understand mechanisms to develop resilient plants.

Keywords: Genomic prediction, high-throughput genotyping, high-throughput phenotyping, phenomic prediction, plant breeding (corn), time-dependent association.

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Introduction

Genetic gain relies on selection intensity, selection accuracy, genetic variance, and the interplay within cycles over time (Heffner *et al.*, 2009). These parameters have direct relationships with both the quality and quantity of phenotype data. Increasing selection intensity and selection accuracy requires growing large numbers of genetically diverse materials and collecting phenotypic data from each genotype across cycles. Traditional phenotyping approaches provide limited data that are resource intensive and scale poorly across time. This results in increased uncertainty and error-prone decision-making with low accuracy in downstream statistical analysis (Furbank and Tester, 2011; Desta and Ortiz, 2014; Araus *et al.*, 2018). Field-based high-throughput phenotyping (FHTP) technology can supply phenomic data that have high time-dimensional (multiple time points) and high organismal resolution, despite increasing computational demands and processing. FHTP is more affordable per observation and less arduous compared with traditional phenotyping methodologies (Shi *et al.*, 2016). Using unmanned (aka unoccupied or uncrewed) aerial vehicles (UAVs, aka drones) equipped with high-tech sensors, plant nurseries and trials can be screened multiple times with high spatial (pixel) and temporal (days) resolution and at different plant growth stages. Reflectance bands captured can be used as novel predictors and/or secondary traits in prediction of agronomically important traits, such as yield and flowering time (Rutkoski *et al.*, 2016; Aguate *et al.*, 2017; Wu *et al.*, 2019; Galán *et al.*, 2020; Adak *et al.*, 2021c; Sandhu *et al.*, 2021). Extracting reflectance band data from remotely sensed images over multiple time points reveals novel temporal phenotypic values for diverse genotypes that can provide key information for plant scientists to discover critical time–trait combinations. This also allows early selection criteria to be proposed, before traditional end-of-season phenotypes, such as grain yield, have been collected (Sun *et al.*, 2019; Krause *et al.*, 2020; Adak *et al.*, 2021c), similar to genomic-based approaches.

Complex quantitative traits (e.g. yield, plant height, and flowering times) generally follow an omnigenic model, where thousands of small interacting loci contribute to the formation of the phenotype (Boyle *et al.*, 2017). To include omnigenic contributions of genome-wide markers in selection, they can be used as predictors of genetic merit for the phenotypic formation of complex traits; this method was first applied in maize as genomic prediction (GP) by Bernardo (1994), also known as genomic best linear unbiased prediction (GBLUP) today. Subsequently, Bayesian GP methods (Meuwissen *et al.*, 2001) and ridge-regression BLUP (rrBLUP) (Whittaker *et al.*, 2000; Endelman, 2011) were developed. Prediction results of these models across genome-wide markers are now applied to the selection of candidate plants known as genomic selection (GS) (Bernardo and Yu, 2007). GS has proved reliable for the selection of complex traits in various crops during plant breeding cycles (Heslot *et al.*, 2015; Windhausen *et al.*, 2012; Zhao *et al.*, 2012; Crossa *et al.*, 2014). However, GS requires significant effort in DNA extractions and genotyping to obtain useful genome-wide markers from populations with GP models. The GS model must be trained for hundreds to thousands of candidate plants, then the model must be updated again in each growing period for selection. To train accurate GS models, large phenotypic datasets must be collected, a bottleneck in prediction. Reducing resources needed and improving the throughput of data collection using FHTP technologies is critical to (i) associate phenotypic traits with genomic data to boost the potential of GS and (ii) use directly as novel predictors, and where a biological basis of these predictors can be explored.

FHTP has revealed dynamic associations between genomic data and temporal phenotypes of complex traits never before possible in plants (Bac-Molenaar *et al.*, 2015; Pauli *et al.*, 2016; Campbell *et al.*, 2017; Feldman *et al.*, 2017; Ward *et al.*, 2019; Anderson *et al.*, 2020; Rodene *et al.*, 2022; Adak *et al.*, 2023b). Phenomic data collected from multiple environments and/or multiple time points can serve as better predictors than genome-wide markers for selecting desired plants in plant breeding (Bernardo, 2021). HTP data can enhance the prediction of complex traits in plants, as well as supply complementary phenomic datasets for genomics to dissect complex traits and their differences in sensitivity to the environments during growth.

When combined with high-revisit temporal UAV phenotypic data, genome-wide markers have had fluctuating phenotypic effects sizes in genome-wide association studies (GWAS) and quantitative trait locus (QTL) studies during crop growth (Anderson *et al.*, 2020; Adak *et al.*, 2021a, 2023b; Wang *et al.*, 2021; Rodene *et al.*, 2022). This phenomenon demonstrates that the same markers can have temporally different fitness for both traits and genotypes. Much like using genome-wide markers as predictors, temporal phenotype data of different sensor-based traits (e.g. vegetation indices) can also be used as predictors. The predictive ability can then be compared with the predictive ability of genome-wide markers (Rincent *et al.*, 2018; Adak *et al.*, 2023b).

Genome-wide markers are collected once, and different individuals are typically straightforward to compare at the same genetic location regardless of when the data are collected. In contrast, phenomic markers are temporally and environmentally dependent and it is still not straightforward to compare phenomic features collected in different environments, at different growth stages, or on different days. Temporal phenotypes are characterized by changing environmental variables across growth stages, resulting in unique trajectories specific to each environment; these unique trajectories may also serve as novel predictors of complex traits across diverse environments. Close similarity between

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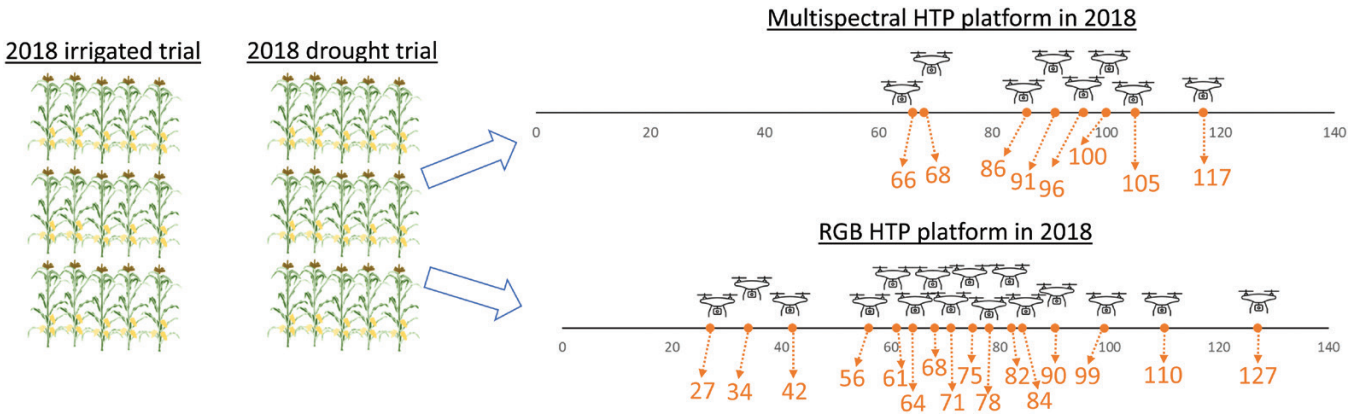


Fig. 1. Field-based high-throughput phenotyping (FHTP) using data from unmanned aerial vehicle (UAV) platforms collected in 2018 irrigated and drought trials. A UAV with an RGB camera was flown in 16 flights throughout growth for the 2018 drought and irrigated trials. A UAV equipped with a multispectral camera was flown in eight flights during 2018 (both irrigated and drought) trials. Timeline of FHTP shows the flight dates as days after planting indicated by orange circles and numbers.

training and testing environments is a requirement for success in genomic prediction; however, this similarity requirement appears lower for temporal phenomic data. Temporal phenomic data capture the variation of different environments and ability to provide more than one different value for each genotype in each environment, which is not possible with genomic data.

This study used recombinant inbred lines (RILs) from the three different bi-parental maize populations totaling 520 RILs, grown in irrigated and non-irrigated (drought) trials that served as different environments in 2018. Multiple UAV surveys were conducted using red–green–blue (RGB) sensors and multispectral sensors in irrigated and drought trials. The 520 RILs were also genotyped using an Infinium genotyping assay. Days to anthesis (DTA), days to silking (DTS), and terminal plant height (PHT) were collected from RILs across irrigated and drought trials and predicted by phenomic and genomic data using the different scenarios. Specifically, our objectives were as follows. (i) To predict the flowering times of unobserved genotypes between irrigated and drought trials using temporal phenomic data derived from RGB and multispectral FHTPs, and compare the phenomic prediction accuracies across irrigated and drought trials. (ii) To compare prediction abilities of different models with combinations of genomic and phenomic data. (iii) To integrate temporal phenomic data with genomic data via a GWAS.

Materials and methods

Developing recombinant inbred lines and experimental design

Parental maize inbred lines used for constructing the three bi-parental linkage populations were developed from crosses Ki3×NC356, LH82×LAMA, and Tx740×NC356 (Chen, 2016). From 2014 to 2018, these populations were advanced to the $F_{7:8}$ using the single seed descent method across College Station (summer nursery) and Weslaco (winter nursery), Texas. In total, 238, 178, and 104 RILs were derived from Ki3×NC356, LH82×LAMA, and Tx740×NC356, respectively. LH82 is a temperate adapted line while Ki3 is tropical and NC356, LAMA, and Tx740 are tropical genetic background but temperate selected lines. The LAMA line [pedigree: (LAMA2002-12-1-B-B-B-B/LAMA2002-1-5-B-B-B-B)-3-2-B-1-B3-B] parent is expected to be 50% related to Tx740 [pedigree: (LAMA2002-12-1-B-B-B)] (Mayfield *et al.*, 2012). All plantings were set using ranges (horizontal grids, 0.76 m between plots) and rows (lateral grids 3.81 m long per plot) accounting for randomized complete block design with two replications. The experimental design is visually represented in Supplementary Fig. S1. The RILs were grown under two different experimental conditions, with and without irrigation, and referred to as irrigated and drought trials in this study. These trials were two different managements at the same location in the same year, but for ease of presentation we use the term environment to separate them here. RILs were planted on 14 March 2018 (irrigated and drought trials). Flowering time includes both anthesis and silking in maize (Colasanti and Muszynski, 2009). DTA and DTS were recorded for each row plot across trials; DTA and DTS were recorded as the time from planting to when at least 50% of a plot initiated anthesis and silking, respectively. PHT was measured manually from the ground to the tip of the tassel in each row plot at the end of the reproductive stage; flowering times and plant heights were recorded in Field book (Rife and Poland, 2014).

Field-based high-throughput phenotyping platforms and image processing

Two different UAV platforms were used in this study. A rotary wing type UAV (DJI Phantom 3 Professional) equipped with an RGB sensor (12 megapixel DJI FC300X camera) was flown at an altitude of 25 m resulting in ~1 cm per pixel resolution. Additionally, a fixed-wing Tuffwing UAV mapper equipped with a multispectral camera (MicaSense RedEdge-MX) was flown at an altitude of 120 m resulting in ~7.5 cm per pixel resolution. In both platforms, 80% side and forward overlap was used. The RGB platform was flown 16 times over irrigated and drought trials. Dates of the flights belonging to both platforms with corresponding days after planting times are shown in Fig. 1.

To obtain the orthomosaics for every flight, raw images belonging to each flight were processed in Pix4Dmapper (https://www.pix4d.com) for RGB, and Agisoft PhotoScan (Agisoft PhotoScan Professional v. 1.2.

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4) for multispectral platforms, respectively. Later, each orthomosaic was subjected to plot-based data extraction.

Data extraction from orthomosaics

A shapefile was first created in QGIS containing the examined population field in each orthomosaic; each orthomosaic was clipped into population levels in QGIS (https://qgis.org/en/site) using the 'Clip Raster by Mask Layer' function, and further data extraction was conducted in R. UAStools (https://github.com/andersst91/UAStools, Anderson and Murray, 2020) and FIELDimageR (https://github.com/OpenDroneMap/FIELDimageR, Matias *et al.*, 2020) packages were used in R to create the shape files and extract the vegetation indices (VIs), respectively. Briefly, (i) shape files for each plot were created using the field map, where range, row, and plot information are stored, using UAStools::plotshpcreate(); (ii) high-resolution orthomosaics (only in the RGB FHTP platform) were reduced using the raster::aggregate(); (iii) FIELDimageR::fieldmask() was used to remove the soil from the vegetation; (iv) FIELDiamgeR::fieldindex() was used to define the VIs; and (v) FIELDimageR::fieldInfo() was used to extract the defined VIs for each plot according to the shapefile created in step (i). More complete descriptions of these steps can be found in (Adak *et al.*, 2023b). VIs used in this study are presented in Supplementary Table S1.

Statistical analysis of phenomic data within an environment

An interaction design was implemented to examine genotypic variation of RILs for each VI extracted from orthomosaics at all flight times of RGB FHTP and multispectral platforms. The interaction design model was used to dissect performance of each RIL across flight times to predict the temporal BLUPs (Adak *et al.*, 2023a). The interaction design was run using restricted maximum likelihood (REML) mixed models for each VI where all components in Equation 1) were attributed as random effects in lme4 in R (Bates *et al.*, 2015). In the following models, a subpopulation effect was not included since no significant differences in variance were found between subpopulations.

$$Y_{ijklm} = \mu + F_i + G_j + (FG)_{(i\times j)} + \text{Range}_k + \text{Row}_l + \text{Rep}_m + \varepsilon_{ijklm} \quad (1)$$

where, Y_{ijklm} is the value of each vegetation index of each row plot at all flight times; μ=overall mean; F_i=the random effect of the ith flight time (as days after planting time; DAP) with $F_i^{iid} \sim N(0,\sigma^2_{Fi})$, $i\in\{66, 68, 86, 91, 96, 100, 105, 117;$ Tuffwing with multispectral camera FHTP platform including irrigated and drought trials$\}$, and $i\in\{27, 34, 42, 56, 61, 64, 68, 71, 75, 78, 82, 84, 90, 99, 110, 127;$ RGB FHTP including irrigated and drought trials$\}$; G_j=the random effect of the jth RIL with $G_j^{iid} \sim N(0,\sigma^2_{Gj})$, $j\in\{1,\ldots,520\}$; $(FG)_{(i\times j)}$=the random effect of the interaction between the ith flight time and jth RIL with $(FG)_{(i\times j)}^{iid} \sim N[0,\sigma^2_{(FG)\ (i\times j)}]$; Range_k=the random effect of kth range with $\text{Range}_k^{iid} \sim N(0,\sigma^2_{\text{Range}k})$, $k\in\{1,\ldots,32\}$; Row_l=the random effect of the lth row with $\text{Row}_l^{iid} \sim N(0,\sigma^2_{\text{Row}l})$, $l\in\{1,\ldots,38\}$; Rep_m=the random effect of the mth replication with $\text{Rep}_m^{iid} \sim N(0,\sigma^2_{\text{Rep}m})$, $m\in\{1,\ldots,2\}$; and ε_{ijklm} is pooled error with $\varepsilon_{ijklm}^{iid} \sim N(0,\sigma^2\varepsilon_{ijklm})$. Results of the interaction between the flight time and RILs $[(FG)_{(i\times j)}]$ component of each VI in Equation 1 was then combined to create the phenomic data and used as predictors in the prediction models. Thus 560 and 704 predictors in RGB and multispectral FHTP, respectively, were extracted to be used in prediction models to predict DTA, DTS, and PHT. A total of 515 RILs (with complete genomic data, below), 16 times, and 35 VIs (32 VIs plus red, green, and blue bands) were used in RGB phenomic data belonging to the drought and irrigated trials. In total, 288 400 data points were used in RGB phenomic data for each trial. A total of 515 RILs (with genomic data), eight times, and 88 VIs (83 VIs plus red, green, blue, red-edge, and near infrared bands) were used in multispectral phenomic data belonging to drought and irrigated trials. In total, 362 560 data points were used in multispectral phenomic data for each trial. Heritability (H) of temporal values of each VI was calculated based on Equation 2:

$$H = \frac{\sigma^2_{G_i}}{\sigma^2_{G_i} + \frac{\sigma^2_{(GF)_{i\times j}}}{\text{no. of flight}} + \frac{\sigma^2{}_{\varepsilon_{ijklm}}}{\text{no. of rep} \times \text{no. of flight}}} \quad (2)$$

Dependent phenotypic traits DTA, DTS, and PHT were predicted to obtain the actual breeding values (ABVs) for each RIL using Equation 1 where the flight components were removed.

Statistical analysis of differences in flowering times and plant height between environments

A genotype by environment model was run for manually collected flowering times (DTA and DTS) and terminal plant height (PHT) combining the irrigated and drought trials based on Equation 3 using REML in mixed model as follows:

$$Y_{ijklm} = \mu + G_i + E_j + (GE)_{i\times j} + \text{Range}_k + \text{Row}_l + \text{Rep}_m + \varepsilon_{ijklm} \quad (3)$$

Where, μ= overall mean; G_i=the random effect of the ith RIL with $G_i^{iid} \sim N(0,\sigma^2_{Gi})$, $i\in\{1,\ldots,520\}$ E_j=the random effect of the jth environment with $E_j^{iid} \sim N(0,\sigma^2_{Ej})$, $j\in\{$irrigated and drought trials$\}$; $(GE)_{(i\times j)}$=the random effect of interaction of the ith RIL at the jth environment; Range_k=the random effect of the kth range with $\text{Range}_k^{iid} \sim N(0,\sigma^2_{\text{Range}k})$, $l\in\{1,\ldots,32;$ irrigated and drought trials$\}$; Row_l=the random effect of the lth row with $\text{Row}_l^{iid} \sim N(0,\sigma^2_{\text{Row}l})$, $l\in\{1,\ldots,38\}$; Rep_m= the random effect of the mth replication with $\text{Rep}_m^{iid} \sim N(0,\sigma^2_{\text{Rep}m})$, $m\in\{1$ and $2\}$; and ε_{ijklm} is pooled error with $\varepsilon_{ijklm}^{iid} \sim N(0,\sigma^2\varepsilon_{ijklm})$.

Heritability (H) for DTA, DTS, and PHT across trials in irrigated and drought trails were calculated based on Equation 4:

$$H = \frac{\sigma^2_{G_i}}{\left(\sigma^2_{G_i} + \frac{\sigma^2_{(GE)_{i\times j}}}{\text{no. of environment}} + \frac{\sigma^2{}_{\varepsilon_{ijkl}}}{\text{no. of environment} \times \text{no. of rep}}\right)} \quad (4)$$

DNA extraction, genotyping, and filtering

DNA procedures were previously documented in Chen (2016) but are briefly described here. Genomic DNA samples extracted from eight seedlings were bulked for each RIL using a CTAB (cetyltrimethylammonium bromide) protocol (Chen and Ronald, 1999). Genomic DNA for each of the 520 genotypes was collected using an Infinium whole-genome genotyping assay in AgReliant Genetics LLC; 515 were successfully genotyped, resulting in 17 444 genome-wide single nucleotide polymorphisms (SNPs). B73 RefGEN_v3 was used to determine the physical location of the genome-wide SNPs (Schnable *et al.*, 2009). Tassel 5 software (Bradbury *et al.*, 2007) was used to filter and impute the genomic data as follows: (i) heterozygote calls set as missing; (ii) markers filtered if allele frequency was <0.05 and missing values were >10% across RILs; and, finally (iii) genomic data were imputed based on LD-kNNi (Money *et al.*, 2015). As a result, 11 334 SNPs were used in GWAS and genomic prediction.

Prediction model and prediction ability

Genomic data and the two different types of phenomics data (RGB and multispectral) were used as predictors in modeling tested and untested

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genotypes in observed and unobserved environments. Flowering times (DTA and DTS) and PHT of 515 RILs were predicted based on the seven different models across irrigated and drought trials using the BGLR package in R (Pérez and de los Campos, 2014). RGB phenomic data contained 288 400 data points (515 RILs×35 VIs×16 flight times) separately for each of the irrigated and drought trials (576 00 data points in total). Multispectral phenomic data contained 362 560 data points (515 RILs×88 VIs×8 flight times) separately for each of the irrigated and drought trials (725 120 data points in total). Seven different models (Table 1) were used to predict DTA, DTS, and PHT across irrigated and drought trials. In these models, y was DTA, DTS, or PHT of 515 RILs grown in irrigated and drought trials; $\boldsymbol{E}$ represented the environment where the genotypes were observed and it was modeled as $E \sim N\left(0, \frac{Z_E Z_E'}{2}\right)$ with Z_E as the incidence matrix that connected phenotypes with environments (irrigated and drought trials); g was the vector of genomic effects and it was modeled using the matrix of SNP markers as linear combinations between p marker and their corresponding effects such that $g = \left\{g_i = \sum_{j=}^{p} x_{ij} b_j\right\} \sim N\left(0, Z_g \mathbf{G} Z_g' \sigma_g^2\right)$ with $G = \frac{XX'}{p}$, $\boldsymbol{X}$ is the centered and standardized (by columns) matrix of marker SNPs (VanRaden, 2008), σ^2_g was the associated variance component, and Z_g was the corresponding incidence matrix that connected phenotypic observations with genotypes; while $\boldsymbol{r}$ and $\boldsymbol{m}$ were the phenomic effects associated with the RGB and the multispectral data, respectively. Similarly to the genomic effect, these were modeled via a linear combination between the VIs derived from FHTP and their corresponding effects such that $r \sim N\left(0, \frac{RR'\sigma_r^2}{16\times35}\right)$ and $m \sim N\left(0, \frac{MM'\sigma_m^2}{88\times8}\right)$ where $\boldsymbol{R}$ and $\boldsymbol{M}$ were relationship matrices derived from RGB and multispectral phenomic data respectively, and σ^2_r and σ^2_m were the associated variance components. The interaction terms between environments E and the genomic markers (g×E) or the VIs derived from RGB FHTP (r×E) or multispectral FHTP (m×E)were modeled according to Jarquín *et al.* (2014). Briefly, the Hadamard product (# cell-by-cell product) of the associated co-variance structures of these model terms was computed such that $g \times E\left(0, \frac{Z_E Z_E' \# Z_g G Z_g' \sigma^2_{g\times E}}{2}\right)$; $r \times E\left(0, \frac{RR' \# Z_E Z_E' \sigma^2_{r\times E}}{16\times35\times2}\right)$, and; $m \times E \sim N\left(0, \frac{MM' \# Z_E Z_E' \sigma^2_{m\times E}}{88\times8\times2}\right)$ with $\sigma^2_{g\times E}$, $\sigma^2_{r\times E}$, and $\sigma^2_{m\times E}$ as the corresponding variance components.

To assess the models, prediction ability was calculated via Pearson's correlation between actual values and predicted values of RILs based on four different scenarios using 5-fold cross-validation with 25 replications; one replication of 5-fold cross-validation is illustrated in Supplementary Fig. S2 and explained as follows (Burgueño *et al.*, 2012). Four prediction scenarios (CV1, CV2, CV0, and CV00) relevant to plant breeding programs were applied to each of the seven prediction models. The tested RILs were randomly selected RILs that were used in the training of the models (4-fold=412 RILs), and the untested RILs were those excluded from the models (1-fold=103 RILs).

Table 1. The prediction models with their notations (× indicates the terms in interaction)

Model	Notation
M1	$y = \mu + E + R + \varepsilon$
M2	$y = \mu + E + M + \varepsilon$
M3	$y = \mu + E + g + \varepsilon$
M4	$y = \mu + E + g + R + \varepsilon$
M5	$y = \mu + E + g + M + \varepsilon$
M6	$y = \mu + E + g + R + M + \varepsilon$
M7	$y = \mu + E + g + R + M + (g \times E) + (R \times E) + (M \times E) + \varepsilon$

For CV1 and CV2, models were trained using the tested RILs in both irrigated and drought trials. Here irrigated and drought trials were considered observed environments together, since predicted (DTA, DTS, and PHT) and predictor variables (phenomic and genomic data) of randomly selected tested RILs from both irrigated and drought trials were used to train models. Then, both the tested and untested RILs were predicted, and the two types of correlations for each model were calculated as follows: the correlation between the predicted and actual values of the untested RILs, called the CV1 scenario, and the correlation between the predicted and actual values of the tested RILs, called the CV2 scenario.

For CV0 and CV00, models were trained using the tested RILs in only the irrigated trial. Here irrigated and drought trials were considered the observed and unobserved environments, respectively, as predicted (DTA, DTS, and PHT) and predictor variables (phenomic and genomic data) of randomly selected tested RILs (4-fold=412 RILs) from only the irrigated trial were used to train models. Then, both the tested and untested RILs were predicted, and the two types of correlations for each model were calculated as follows: the correlation between the predicted and actual values of the tested RILs (the same 412 RILs trained in the irrigated trial) in the drought trial was called the CV0 scenario. The correlation between the predicted and actual values of the untested RILs (the remaining 103 RILs) in the drought trial was called the CV00 scenario. To sum up, CV1 indicates the prediction of untested RILs in an observed environment; CV2 indicates the prediction of tested RILs in an observed environment; CV0 indicated the prediction of tested RILs in an unobserved environment; and CV00 indicates the prediction of untested RILs in an unobserved environment. Prediction was performed from favorable to unfavorable environments in CV0 and CV00 scenarios.

The average prediction ability (r_φ) was calculated based on Tiezzi *et al.* (2017) using the prediction ability of each environment and its sample sizes.

$$r_\varphi = \frac{\sum_{i=1}^{I} \frac{r_i}{V(r_i)}}{\sum_{i=1}^{I} \frac{1}{V(r_i)}}$$

Where, r_i is Pearson's correlation between actual values and predicted values in the ith environment, $V(r_i) = \frac{1-r_i}{n_i-2}$ is the sampling variance, with n_i being the number of RILs in an environment.

Temporal genomic prediction

Temporal values of the NDRBI (normalized difference red blue index) from high-resolution RGB phenomic data were used; this model was intended to examine the prediction ability of genome-wide markers for temporal values of this trait derived from RGB FHTP. The temporal genomic prediction model was used as follows:

$$y = \mu + F + g + (g \times F) + \varepsilon \quad (5)$$

Here y is the NDRBI values of RILs (515) derived from Equation 1 belonging to 16 time points (27, 34, 42, 56, 61, 64, 68, 71, 75, 78, 82, 84, 90, 99, 110, and 127 DAPs); F represents the flight effect and it is modeled via Z_F, the incidence matrix that accounts for 16 flights used in RGB FHTP; and g is the genomic component modeled as previously described (VanRaden, 2008). The interaction (g×F) is between

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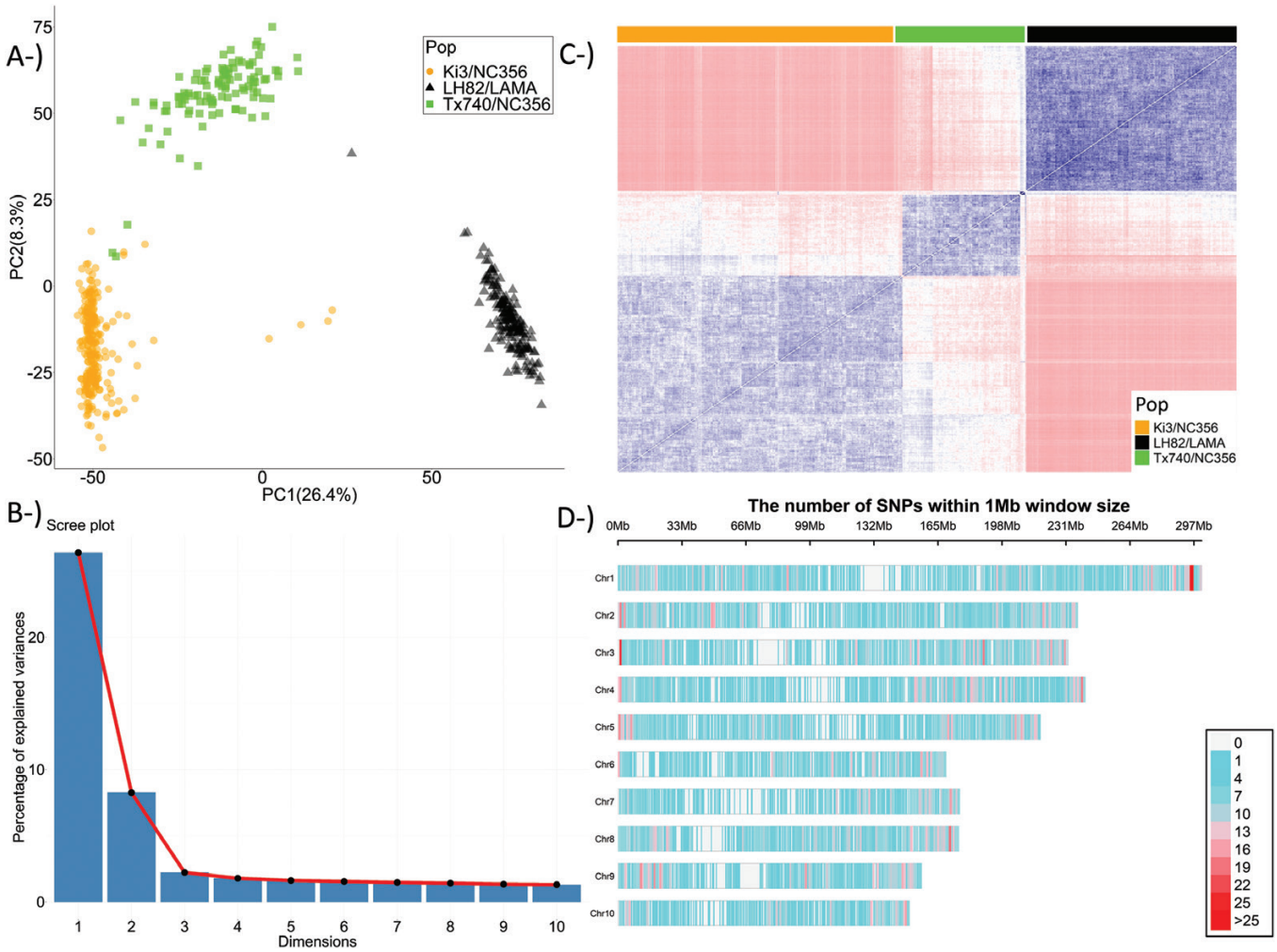


Fig. 2. (A) Biplot illustration including the first two principal components (PCs) showing separation of 520 recombinant inbred lines (RILs). (B) Scree plot showing explained percentage variation by each PC. (C) Distance matrix showing the population structure of 520 RILs. (D) Distribution of 11 334 filtered SNPs across 10 chromosomes.

the genomic relationship matrix and the flight matrix and it is modeled such that $g \times F \sim \left(0, \frac{Z_F Z_F{}' \# Z_g \mathbf{G} Z_g' \sigma^2_{g\times F}}{16}\right)$. This model was run for each trial (drought and irrigated) separately. Prediction was performed similarly to as described in 'Prediction model and prediction ability'. Briefly, 5-fold cross-validation with 25 replications was applied; the model was trained using tested RILs (4-fold=412 RILs) and the remaining RILs (1-fold=103 RILs) were untested RILs that were used to calculate the prediction ability in each replication. Pearson's correlation was calculated as a unit of prediction ability between predicted and actual values of NDRBI of untested RILs for each trial and time point. The Wilcoxon test was applied to compare the prediction ability between trials at each time point.

Genome-wide association study of temporal traits and flowering times

Beyond prediction, it is valuable to understand the underlying biological function of phenomic data that predict the DTA, DTS, and PHT. The GLM (general linear model) with three principal components (PCAs) (GLM+PCA) was conducted in GAPIT (version 3) for GWAS in R using 11 334 filtered SNPs (Wang and Zhang, 2021). Predicted genotypic effects of DTA, DTS, and PHT and predicted phenotypic effects of all VIs belonging to all flight times in RGB and multispectral FHTP platforms belonging to the irrigated and drought trials were used as phenotype data. Overall, 35 VIs belonging to 16 flight times derived from RGB FHTP and 88 VIs belonging to eight flight times derived from multispectral FHTP collected across 515 RILs grown in the irrigated and drought trials were used as phenotype data in GWAS. PCA was calculated from genomic data in Tassel 5 software (Bradbury *et al.*, 2007). Three PCs (explaining 26.4, 8.3, and 2.2% among the first three PCs) were used to account for population structure of the three RIL populations. Population structure, PC scores, and SNP distribution across each chromosome are shown in Fig. 2. A Bonferroni threshold of 5.36 was calculated based on a *P*-value of 0.05 (0.05/no. of markers); this threshold was applied to consider the SNPs as significant. Meff significance threshold was calculated as 4.48 (Li and Ji, 2005). Candidate genes were identified within 100 kb of discovered SNPs, as well as within the linkage disequilibrium (LD) block where the minimum R^2 value was 0.8. LD was calculated and visualized using the LDheatmap package in R (Shin *et al.*, 2006). LD blocks of 10 chromosome are presented in Supplementary Fig. S3.

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Results

Interaction design results: temporal repeatability, variance components, temporal breeding values, and correlation

The RGB platform had higher ground resolution (~1 cm per pixel) than the multispectral platform (~7.5 cm per pixel), indicating that RGB had greater resolution. Among the major causes was a lower flight altitude (25 m) of the RGB platform than the multispectral HTP platform. This resulted in overall higher heritability values in RGB phenomic data. Heritability of VIs in multispectral phenomic data varied between ~0.10 and 0.85 in the irrigated trial, and ~0.13 and 0.86 in the drought trial, whereas heritability of VIs in RGB phenomic data varied between ~0.84 and 0.93 in the drought trial, and ~0.86 and 0.92 in the irrigated trial (Fig. 3). The flight component in the interaction design (Equation 1) explained the highest total variation for most VIs in the RGB phenomic dataset (Fig. 3). Results of interaction terms [$(FG)_{(i\times j)}$ in Equation 1], in which genetic materials (RILs) were considered to have interactions across flights, were used to predict temporal breeding values of RILs for each VI in RGB and multispectral phenomic data. Correlations between individuals VIs separately in drought and irrigated phenomic data and flowering time and plant height were found to change temporally between –0.4 and +0.4 depending on flight dates. Temporal RGB VIs had a greater magnitude of correlations in flowering times from positive to negative than multispectral VIs (Fig. 4).

Phenomic data correlations showed that VIs belonging to the same growth stages were usually correlated with flowering and plant height in a similar way (Fig. 4). Temporal data of VIs had minimal multicollinearity in the phenomic dataset (Fig. 5).

Differences in flowering times and plant height between different environments

The genotype by environment model (Equation 3) revealed differences in flowering times and plant heights across irrigated and drought trials. Interaction variance components between genotype (RILs) and environments explained ~8% of total variation for DTA and DTS, and 19% for plant height (Fig. 6). The changes in flowering times and plant height across different environments were used as dependent predicted variables to assess predictive ability of genomic, phenomic, and combined data in applying four different prediction scenarios.

Results of prediction scenarios

In all prediction scenarios, models containing phenomic and genomic data (M4, M5, M6, and M7) predicted the flowering times and plant height numerically better, and generally significantly better, than models using only genomic data (M3) (Fig. 7). This led to the highest increases in prediction ability related to untested genotypes in observed and unobserved environments (CV1 and CV00 scenarios). In scenario CV1, flowering times (DTA and DTS) and plant height (PHT) were predicted with ~0.58, 0.59, and ~0.42 prediction ability using only genomic data (M3). However, prediction ability was further increased up to 0.75 to 0.56, respectively, for flowering times and plant height using genomic and RGB phenomic data together (M4) (Fig. 7). Similarly, prediction ability was

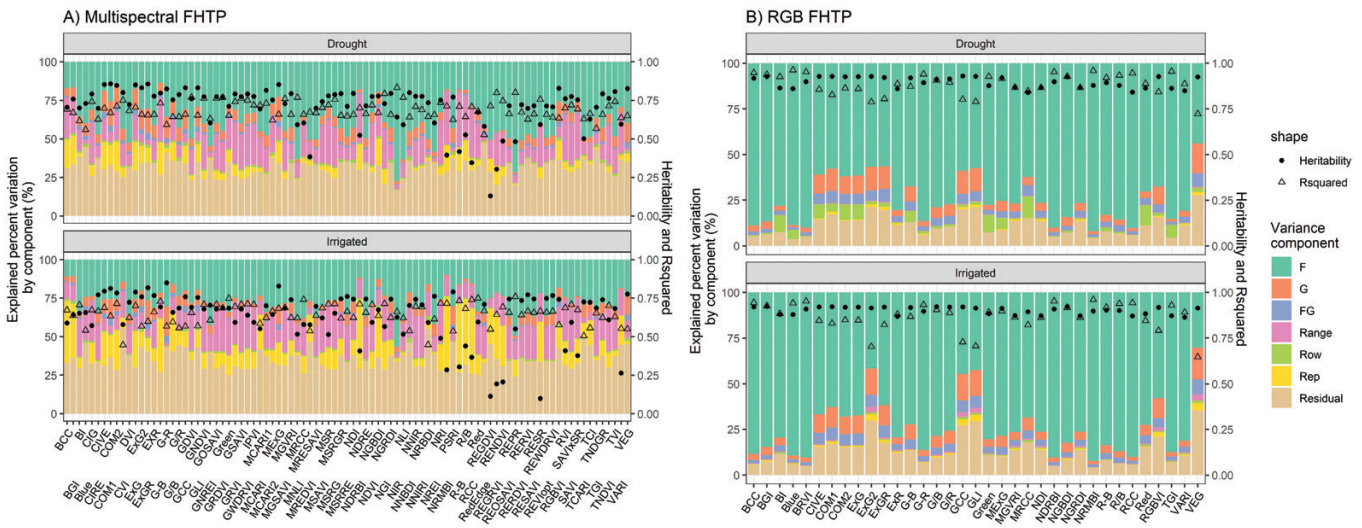


Fig. 3. Stack bar plot shows the explained percentage variation by each component in the interaction design (Equation 1) for the RGB and multispectral phenomic data in 2018. The left *y*-axis shows the explained percentage variation of each component in the model, the right *y*-axis show the R^2 (filled circles) and heritability (open triangles). The *x*-axes show the vegetation indices. (A) Results of the multispectral data for drought (above) and irrigated (below) trials in 2018. (B) Results of the RGB phenomic data for drought (above) and irrigated (below) trials in 2018.

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Fig. 4. Correlations as heatmap illustration between FHTP phenomic data and phenological traits (days to anthesis, DTA; days to silking, DTS; and terminal plant height, PHT). Vegetation indices (on the *y*-axis) belonging to each time point (on the lower *x*-axis as days after planting) show the FHTP phenomic data. DTA, DTS, and PHT phenological traits are on the upper *x*-axis. (A) Correlations with RGB and (B) correlations with multispectral phenomic data. Pearson's correlation was used.

increased up to 0.67 (DTA and DTS) and to 0.52 (PHT) for flowering times and plant height using genomic and multispectral phenomic data together (M5) (Fig. 7). Slightly higher prediction abilities in CV1 were achieved by including interactions for flowering times (M7; 0.77 for DTA and 0.76 for DTS) and plant height (M7; 0.58) than without (M6) (Fig. 7). In scenario CV00, model (M6) with genomic and both RGB and multispectral phenomic data predicted flowering times and plant height consistently better than other models (M1–M5). More importantly, M6 predicted the flowering times and plant height 0.13 and 0.14 better than M3 using only genomic data (Fig. 7).

Genome-wide association results

The number of significant SNPs discovered was greater in the irrigated trial than in the drought trial using both RGB and multispectral platforms; NDRBI and RCC (red chromatic coordinate index) provided the highest number of SNPs in RGB and multispectral FHTP, respectively (Fig. 8A). It was found that the density of SNPs discovered was more intense during flowering time in RGB phenomic data than at other times; however, the density of SNPs discovered was found to be close over time in the multispectral phenomic data (Fig. 8B). Combining all phenotypic data across environments, temporal multispectral VIs included 704 phenotypes in irrigated and drought trials; and temporal RGB VIs included 560 phenotypes in irrigated and drought trials. These 2528 temporal×VI phenomic measures resulted in significant associations of 2954 SNPs (Supplementary Table S2), of which 1394 SNPs were unique to RGB FHTP, 442 SNPs were unique to multispectral FHTP, and 1117 SNPs were commonly discovered in both FHTP (Fig. 8C). GWAS hits were represented as Manhattan plots for DTA, DTS, and PHT (Fig. 9A), RGB (Fig. 9B), and multispectral phenomic data (Fig. 9C; only SNPs above the significance threshold are shown in Fig. 9B and C).

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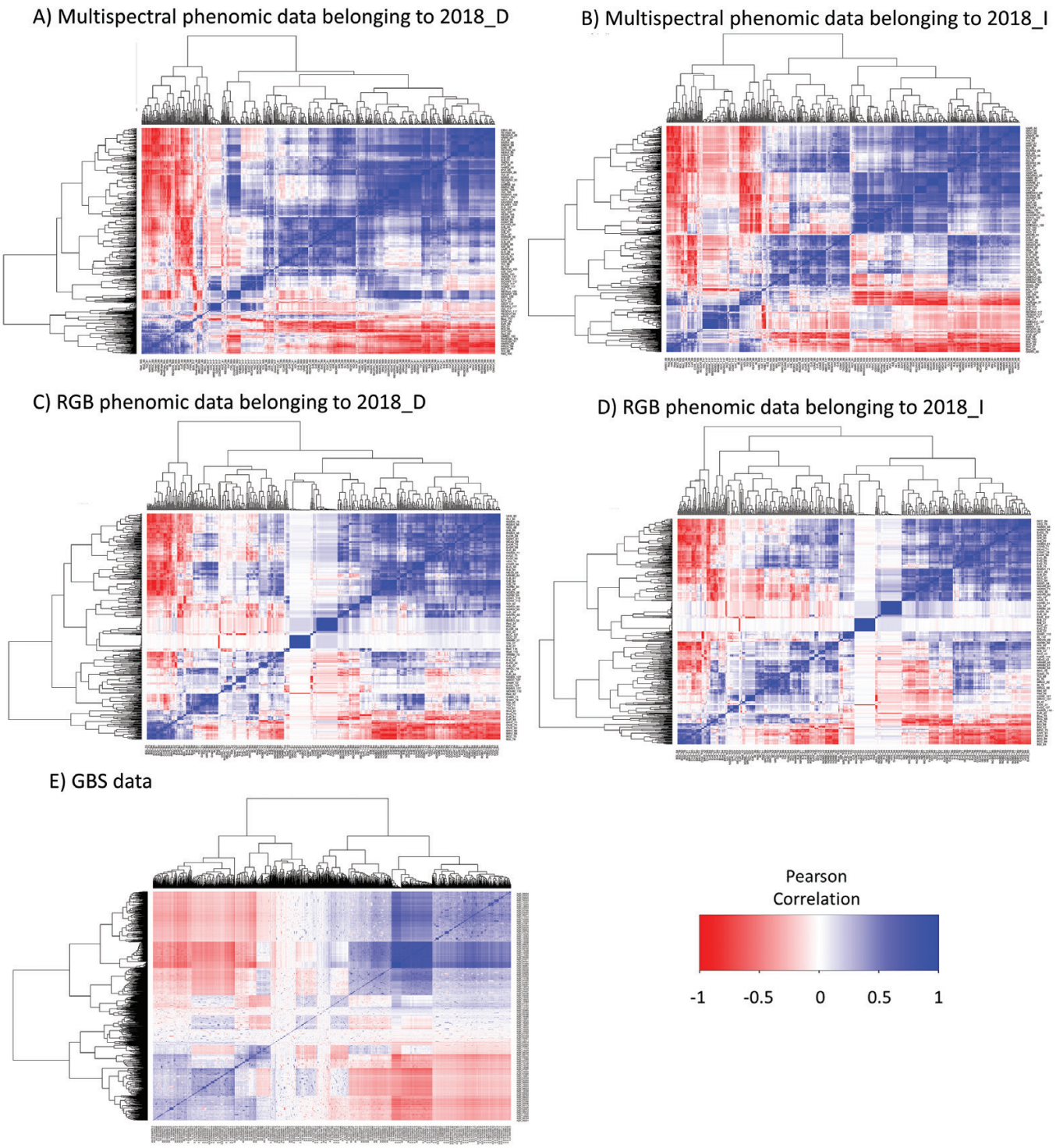


Fig. 5. Correlation of temporal VIs using (A) multispectral phenomic data in the 2018 drought trial (2018_D) and (B) the irrigated trial (2018_I), (C) RGB phenomic data in the 2018 drought trial (2018_D) and (D) the irrigated trial (2018_I). (E) GBS data.

Revealing interplay between genetic and environmental changes by FHTP

Temporal blue index values belonging to between 61 and 82 DAPs led the discovery in GWAS (Supplementary Table S2; Fig. 9) of an SNP (chr3: base pair 8 888 783) in the genic region of the *hsp18f* candidate gene (chr3: base pairs 8 886 055 to 8 890 247), a member of the heat shock protein (HSP) family in maize, in both irrigated and drought trials. It is interesting, however, that temporal blue index values belonging to

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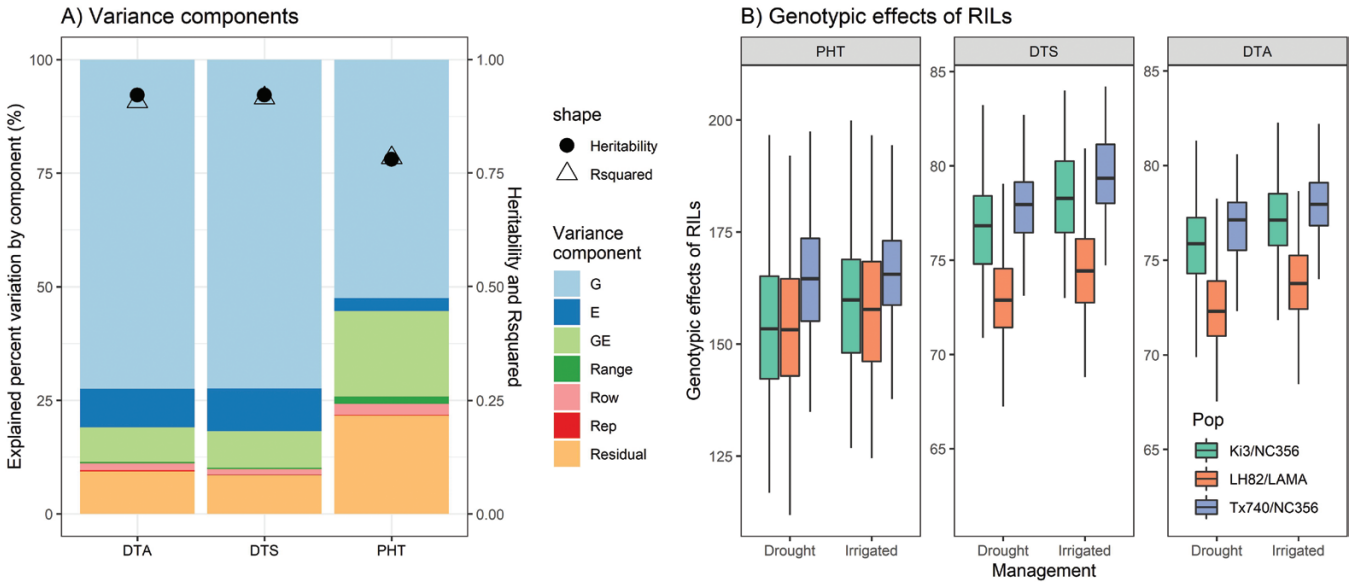


Fig. 6. Variance components of the genotype by environment model (Equation 3). (A) Explained percentage variation by each component in Equation 3 for flowering times (DTA and DTS) and terminal plant height (PHT). (B) Box plots of the best linear unbiased predictors of DTA, DTS, and PHT of recombinant inbred lines belonging to their subpopulations across trials including the genotype by environment interaction component $[(GE)_{(i\times j)}]$ in Equation 3.

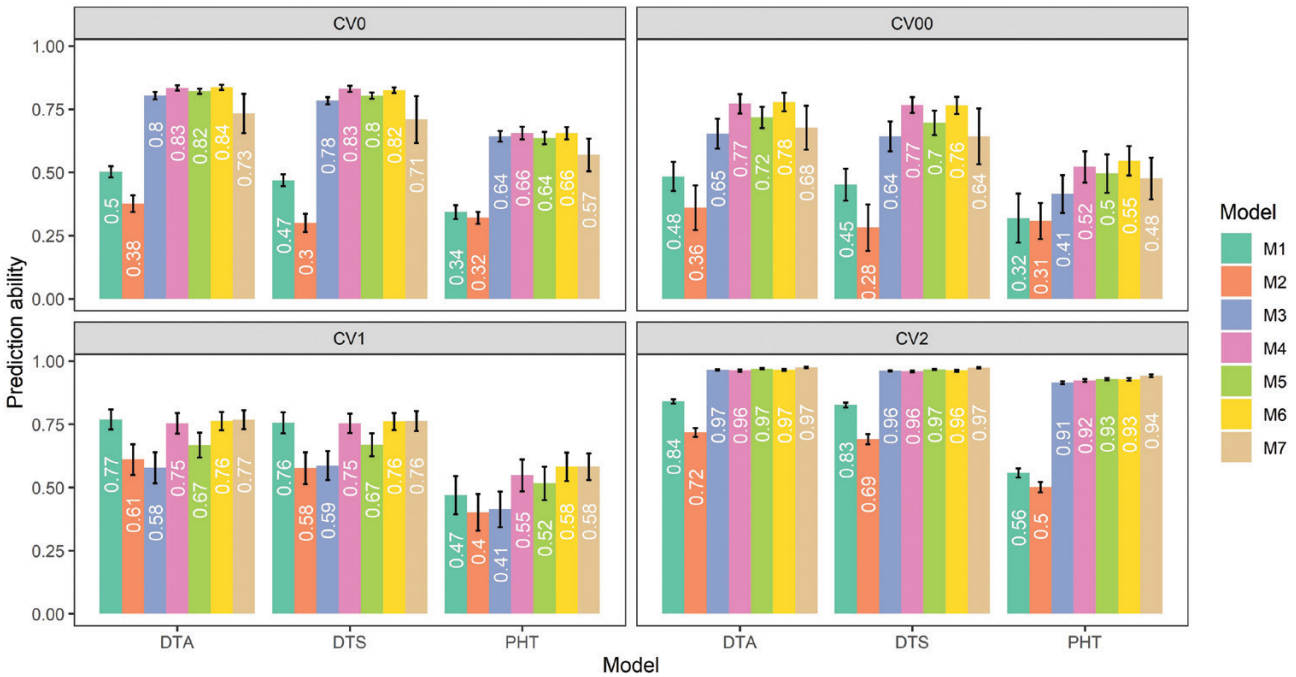


Fig. 7. Prediction ability of the seven models (M1–M7; Table 1) based on four prediction scenarios for three traits. CV00 and CV0 indicate prediction scenarios of untested and tested genotypes in the unobserved environment, respectively. CV1 and CV2 indicate prediction scenarios of untested and tested genotypes in observed environments, respectively. Prediction ability (*y*-axis) is the Pearson's correlation between predicted and actual value of RILs depending on the CVs.

flight times before or after 61 or 82 DAPs, respectively, did not discover this locus (Fig. 10A). The time period from 61 to 82 DAPs contains the DTA and DTS flowering, and the temperature (min, average, and max) was highest (°C) during flowering (Fig. 10A). Allelic variation of discovered loci in GWAS (chr3: base pair 8 888783) linked to *hsp18f* caused statistically significant differences for plant height and flowering time across RILs (Fig. 10B).

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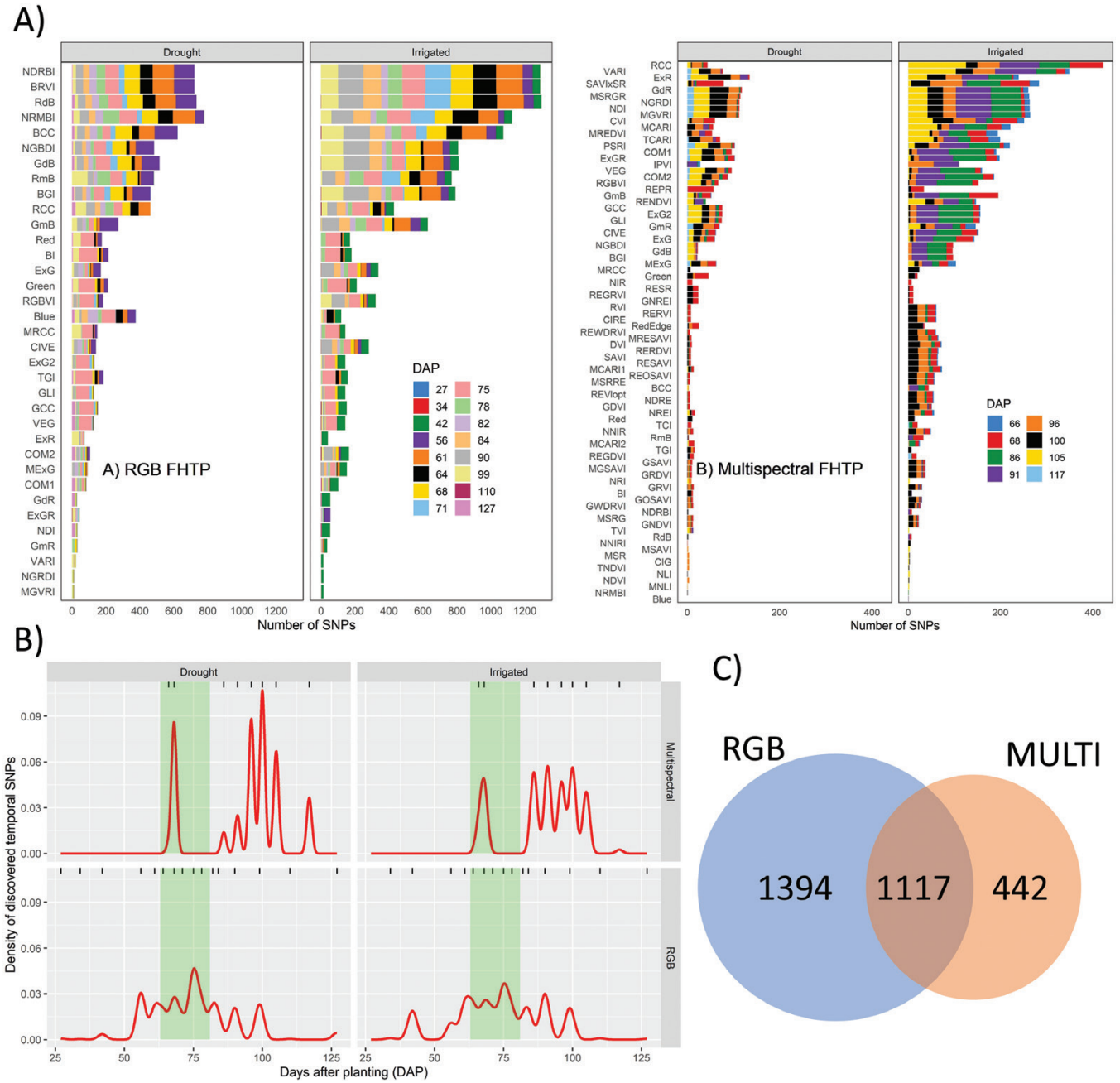


Fig. 8. (A) Number of SNPs discovered by using the temporal VIs belonging to RGB (left) and multispectral (right) FHTP phenomic data in 2018 drought and irrigated trials by flight date. (B) Density plot of significant temporal SNPs across flight times based on RGB (top) and multispectral (bottom) FHTP as well as drought (left) and irrigated (right) environments. (C) Venn diagram showing the unique and common SNPs discovered for RGB and multispectral phenomic data across both trials.

Temporal genomic prediction results

Temporal genomic prediction by Equation 5 resulted in variation in prediction ability of temporal values of NDRBI (Fig. 11A), and the flight effect (F) explained the highest variation, while interactions between RILs and flights ($g \times F$) were found to be low (Fig. 11B). Temporal NDRBI values belonging to time points between 75 and 99 DAPs were genomically predicted to be greater than 27–71 DAPs in both trials (Fig. 11). Temporal NDRBI values belonging to the drought trial had a statistically greater genomic prediction ability than those in the irrigated trial during earlier growth times (e.g. 27, 34, 56, 61, and 68 DAPs) (Fig. 11). Conversely, temporal NDRBI values belonging to the irrigated trial had a statistically greater

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Fig. 9. (A) Manhattan plots for flowering times and terminal plant height in drought and irrigated trials in 2018. The four loci highlighted are further described in the Discussion. (B) Manhattan plots for multispectral phenomic data (88 VIs×8 flight times×2 two trials: 1408 phenotypes), with only statistically significant SNPs shown. (C) Manhattan plots for RGB phenomic data (35 VIs×16 flight times×2 trials: 1120 phenotypes), with only statistically significant SNPs shown. (D) LD regions around the four loci.

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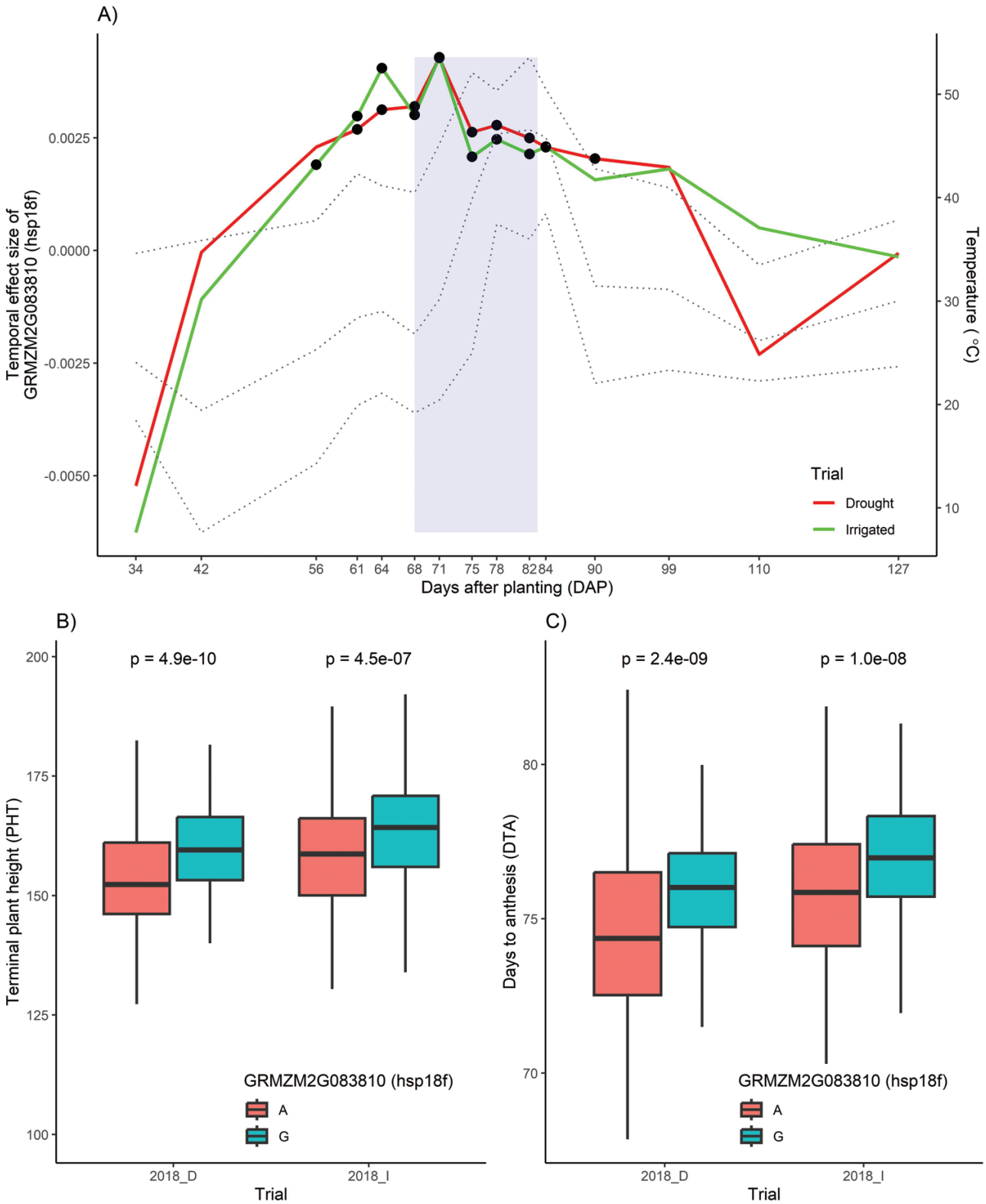


Fig. 10. (A) The left *y*-axis shows temporal effect size for using the blue index of *GRMZM2G083810* (*hsp18f*) in drought and irrigated trials in 2018. *GRMZM2G083810* (*hsp18f*) was significant in GWAS using the blue index belonging to flights indicated by filled circles. The right *y*-axis provides temperatures (°C) for the three dashed lines showing the highest, average, and lowest (from above to below) temperature belonging to each DAP on the *x*-axis. Allelic segregation (adenine, A; and guanine, G) belonging to *GRMZM2G083810 (hsp18f)* differentiated haplotypes with A versus G in terms of terminal plant height (B) and days to anthesis (C). Haplotype groups were compared using Wilcoxon test, and *P*-values are located above the boxplots.

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prediction ability than those in the drought trial during later growth times (e.g. 90, 99, and 110 DAPs) (Fig. 11). Temporal marker effects were found to be dynamic, contingent upon NDRBI values belonging to flight times for each locus (Fig. 12). Effects of genomic markers could change from negative to positive or from positive to negative (Fig. 12).

Discussion

Three GWAS hits of note were discovered for flowering traits (DTA and DTS) in both irrigated and drought trials (Fig. 9A). Two of those (*flw_1* and *flw_2*) are in the fifth bin location in chromosome 8; this bin location was previously described as a hot spot for flowering times in maize (Koester *et al.*, 1993; Chardon *et al.*, 2004; Salvi *et al.*, 2007). *Flw_1* (chr8: base pair 138 861 176) and adjacent loci with least –log(*P*) >4.48 were discovered between base pairs 131 779 714 and 144 732 487 in chromosome 8 for flowering times (Fig. 9A; Supplementary Table S2); this region contains the *ZmRap2.7* AP2/EREB domain transcription factor (chr8: base pairs 135 654 097–135 657 524) that is related to the *vegetative to generative transition* (*vgt1*) gene controlling the flowering time in maize (Salvi *et al.*, 2007). *Flw_2* (chr8: base pair 124 457 676) and adjacent loci

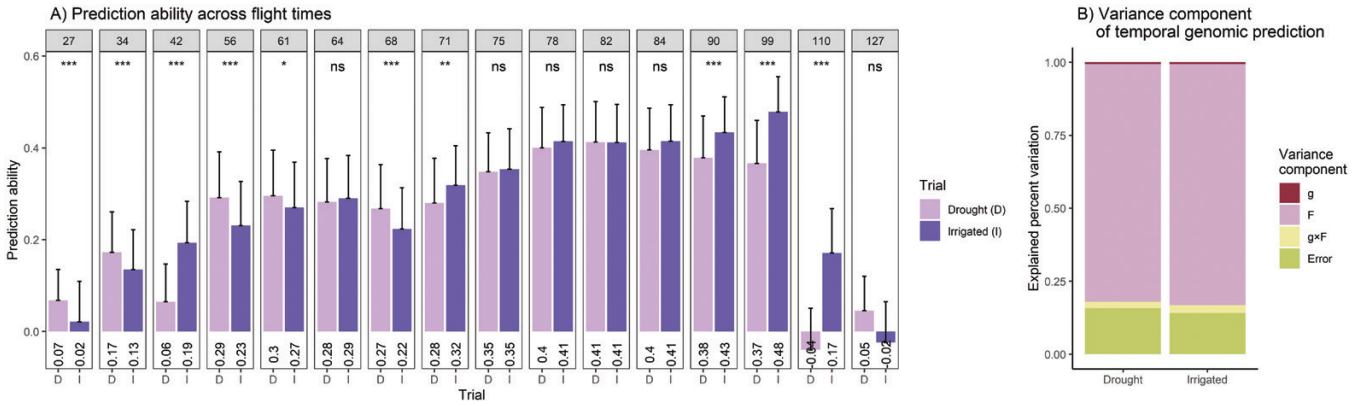


Fig. 11. (A) Temporal genomic prediction ability results for NDRBI vegetation index belonging to drought and irrigated trials based on the temporal genomic prediction model (Equation 5). (B) Explained percentage variation by the components in Equation 5. Temporal genomic prediction abilities were found to be significantly different between irrigated and drought trials at multiple time points based on Wilcoxon comparisons at the top of each flight time. *, **, *** are significant at 0.05, 0.01, and 0.001, respectively; ns is not significant. The mean prediction ability was placed at the bottom of each bar plot.

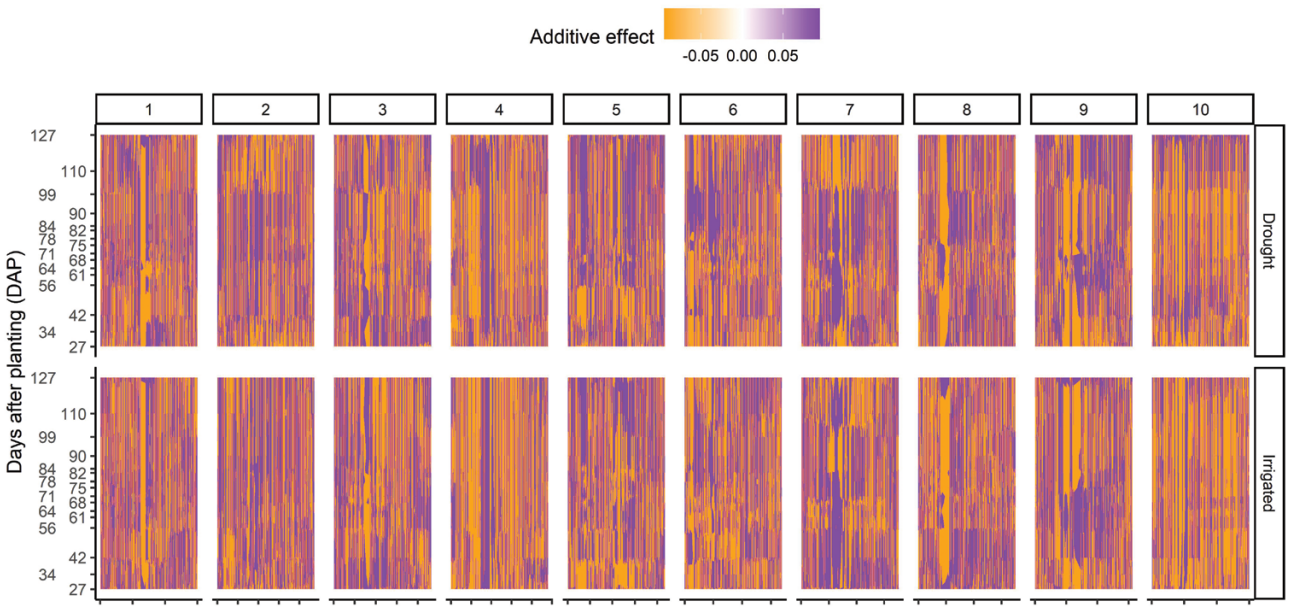


Fig. 12. Temporal additive marker effects for NDRBI across chromosomes (on *x*-axis) and flight dates as days after plantings (on *y*-axis; DAP) in irrigated and drought trials in 2018.

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with at least –log(*P*) >4.48 were discovered between base pairs 120 455 777 and 128 336 449 in chromosome 8 for flowering times (Fig. 9A; Supplementary Table S2); this region contains *ZCN8* (chr8: base pairs 126 680 822–126 680 822) that promotes the transition from vegetative to generative phase encoding the florigen hormone in maize (Meng *et al.*, 2011). *Flw_3* (chr1: base pair 271 627 282) and adjacent loci with at least –log(*P*) >4.48 were discovered between base pairs 271 627 282 and 280 637 432 in chromosome 1 (Fig. 9A; Supplementary Table S2). This region contains *Dwarf8* (*D8*) (chr1: base pairs 272 696 026–272 696 147) that impacts flowering time variation as well as plant height in maize (Thornsberry *et al.*, 2001). *Flw_2* and its adjacent loci with at least –log(*P*) >4.48 containing *ZCN8* were discovered by various VIs belonging to mostly 68 DAP in multispectral FHTP and 42 DAP in RGB FHTP (Fig. 9A–C; Supplementary Table S2). Similarly, *Flw_1* and its adjacent loci containing *ZmRap2.7* were similarly discovered around flowering times by the VIs in multispectral data belonging to 68 DAP (Fig. 9A–C; Supplementary Table 2). Furthermore, *Flw_3* and its adjacent loci containing the *Dwarf8* gene were discovered by VIs in RGB FHTP belonging mostly to 82 DAP (Fig. 9A–C; Supplementary Table S2). Overall, these results indicate that VIs derived from remotely sensed images can capture known variation in flowering time and discover flowering-related loci in association mapping, in most cases even earlier than using actual flowering times. Because temporal values of VIs are capable of detecting flowering time variation even before flowering, their early and temporal assessment of complex traits could help decipher the underlying causal biology and growth modeling (Anderson *et al.*, 2019).

Any large phenomic dataset raises concerns about multicollinearity, redundancy in predictors, and potential resulting problems, especially model overfitting (Adak *et al.*, 2021c). The temporal phenomic and genomic data were subjected to correlation calculations to assess multicollinearity, both ranging from –1 to 1 (Fig. 5). We found no evidence that multicollinearity was any more present in temporal phenomic data than in genomic data (Fig. 5), despite having a relatively sparse genomic marker set compared with many modern studies. This suggests that if the scientific community is not concerned about multicollinearity in genomics, there should be even fewer concerns in phenomics at this resolution. Regardless, modern machine learning minimizes the major concerns around multicollinearity analysis (Montesinos López *et al.*, 2022). Both the phenomic and genomic measures remained unsaturated in this study, and more measures (more features in phenomics, more SNPs in genomics) could prove beneficial to improve predictions.

Among phenomic data, the majority of observed multicollinearity was between some of the many VIs calculated within a given time point. However, it remained surprising how few multicollinearity issues there were given the limited number of spectral bands used to calculate VIs. This suggests that there might be additional VIs that could be utilized to improve prediction. Essentially no additional resources are needed to calculate additional VI measures within a UAV time point, so it is logical to extract as many VI features as possible from each orthomosaic, even if some multicollinearity is observed. More noteworthy was the low multicollinearity across the relatively high temporal dimension (multiple flight times near to each other). This suggests that measurable and unique biological variation occurs within very short windows of time. A higher temporal dimension and phenomic features could be used to both improve prediction and to better understand plant development. Various FHTP-related predictions reported inconsistency across different years and environments (Rutkoski *et al.*, 2016; Crain *et al.*, 2018) and this may stem from using low number of phenomic feature (VIs) or low temporal flight resolution. Conducting additional UAV flights requires resources proportional to the number of flights, so finding the optimal number and times to collect data remains an important open question in FHTP.

Temporal phenomic data used as predictors in prediction models generated promising prediction accuracies between diverse trials. Phenomic tools have appeared to be as good or better than genomic tools (Rincent *et al.*, 2018; Adak *et al.*, 2023a). Similarly, we have found that RGB phenomic data predicted flowering times and plant height better than genomic data (M1 versus M3 in CV1, Fig. 7). However, we also found that combined data (phenomic plus genomic data) with interactions with environment yielded greater predictive ability than using genomic or phenomic data alone for flowering time and plant height in maize (M4 to M7 in Fig. 7). Thus, temporal phenomic data add complementary predictive information for flowering time and perhaps other complex traits in maize.

Temporal phenomic prediction of complex traits

In previous studies, agronomic complex traits (e.g. flowering time and yield) were predicted from one or a limited number of VIs belonging to one or a limited number of time points and in one environment. For example, BNDVI (normalized difference between the blue and near infrared wavelength bands) was used as a predictor to predict grain yield and flowering time in maize using the PLS regression; the prediction accuracies (0.54–0.79) varied depending on the temporal resolution (number of flights) (Wu *et al.*, 2019). Prediction accuracies for flowering times as well as grain yield increased across diverse environments when a higher number of VIs over multiple time points were jointly used (Adak *et al.*, 2021c, 2023a). Aguate *et al.* (2017) evaluated the performance of 62 hyperspectral reflectance bands between 392 nm and 850 nm obtained at 300 m of flight elevation in predicting maize yield across diverse scenarios (populations, small plots, a farmers field, etc.) and over five time points. They found that single time point-derived reflectance

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bands predicted ~10–20% lower yield than using all five time points, with their highest prediction accuracies reaching up to ~0.49%. However, when comparing separate studies, differences in populations and environments measured cannot be ignored. Considering resources needed, the present study supports that high-resolution images derived from low-altitude RGB belonging to a higher number of time points can provide better prediction than a higher altitude with more spectral bands belonging to fewer time points (M1 versus M2 in Fig. 7). Overall, both higher pixel resolution (~1 cm of ground per pixel) that comes from lower flight elevations or improved camera resolution, along with more frequent flights, shows that there is potential for higher prediction accuracies (>0.5) for grain yield in maize in this and previous studies (Adak *et al.*, 2021c, 2023a). Adding additional spectral bands (e.g. hyperspectral) without compromising resolution or flight frequency might add additional benefits.

Genomic and phenomic prediction

CV1 and CV00 demonstrated that the predictive abilities of genome-wide makers decrease when it comes to predicting unknown genotypes in known and unknown environments (Fig. 7). This is in part because differences in flowering time driven by different environments cannot easily be taken into consideration by genome-wide markers, since genomic variation remains fixed across all environments (Jarquín *et al.*, 2014; Cooper *et al.*, 2016). It is therefore logical that results of CV1 showed that prediction of flowering times and plant height of unknown genotypes in known environments was better predicted with high-resolution RGB phenomic data than with either genomic or multispectral phenomic data alone (M1 versus M2 and M3 in CV1; Fig. 7). Combining phenomic and genomic data in prediction better captures the genomic and environmental variation together, especially in the most challenging scenario, namely the prediction of the performance of untested genotype in untested environments (CV00).

Boosting prediction ability by combining VIs with genomic data has also been shown for yield in wheat and rye (Rutkoski *et al.*, 2016; Crain *et al.*, 2018; Sun *et al.*, 2019; Galán *et al.*, 2020). Our results support that instead of using single pre-selected VIs (as is common) or limited reflectance bands, a higher number of VIs may provide greater prediction ability and high consistency across environments without issues of additional cost or multicollinearity. It is notable that this is a phenomic parallel to a similar transition trend from GWAS and marker-assisted selection, targeting specific markers, to genomic prediction using all markers.

Integration of high-throughput phenomic data with association mapping and genomic prediction

Temporal plant height (aka, canopy height measurement; CHM) captured by UAV has previously been used as phenotype data in association mapping in maize hybrids (Adak *et al.*, 2021b, 2023a) and inbreds (Wang *et al.*, 2019; Anderson *et al.*, 2020), resulting in discovery of numerous temporal loci. Temporal loci explain phenotypic variation of plant height differently depending on the specific growth stage; this allows better biological understanding of the loci segregating in a population that control different stages of growth. Similar findings have been shown in soybean using UAV-based phenotyping (Xavier *et al.*, 2017), in cotton using ground-based vehicle phenotyping (Pauli *et al.*, 2016), in wheat, rice, and triticale using tractor-based phenotyping (Würschum *et al.*, 2014; Tanger *et al.*, 2017; Lyra *et al.*, 2020), as well as in maize and barley using automated-based phenotyping under greenhouse conditions (Muraya *et al.*, 2017; Neumann *et al.*, 2017; Zhang *et al.*, 2017). Temporal NDVIs (normalized difference vegetation indexes) captured by UAVs have also been used in association studies in cotton (Pauli *et al.*, 2016), sorghum (Spindel *et al.*, 2018), and maize (Wang *et al.*, 2021). However, the present study showed that NDVI discovered among the lowest number of SNPs in GWAS mapping (Fig. 8A). This reinforces that using only one or a few temporal phenotypes of VIs or CHM will not broaden the discovery of segregating temporal loci controlling plant growth.

It was intriguing that high-resolution RGB images discovered more loci during mid-growth, around flowering, than other growth periods, while low-resolution multispectral images discovered loci quite close during and after flowering times (Fig. 8B). It seems likely that the high-resolution RGB better captured the differences in plant development such as leaf area index, and senescence with more plant pixels. It is likely that multispectral images better distinguished tassel pixels from leaf pixels during flowering than RGB. These results showed that temporal phenotypes captured by multiple drone surveys and sensors can discover previously unrevealed time-dependent associations between temporal traits and genome-wide polymorphisms (Fig. 9; Supplementary Table S2). Overall, dissection of temporal plant growth is better if high time-dimensional phenotype data are used in association mapping studies.

Validation of a candidate gene using temporal phenotype data

A heat stress-related candidate gene, *GRMZM2G083810* (*hsp18f*), was discovered using the temporal blue index phenotype around only flowering time in both drought and irrigated trials. Flowering time is among the most sensitive periods to heat and drought stress (Barnabás *et al.*, 2008), and environmental heat stress probably led to the discovery of *GRMZM2G083810* (*hsp18f*) during flowering (Fig. 10A). This finding demonstrates that temporal phenotypes can enhance discovering opportunistic (oQTLs) (Wu *et al.*, 2004). High-resolution HTFP images in particular can capture the specific interplays between plant growth and biotic stresses during

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certain growth periods. The *GRMZM2G083810* (*hsp18f*) candidate gene caused the RIL haplotypes to have statistically different flowering as well as terminal plant height (Fig. 10B, C) in both drought and irrigated trials. It is also surprising that *GRMZM2G083810* (*hsp18f*) was discovered for plant height in the drought trial (Fig. 9A). This indicated that the allelic variation associated with heat stress-related genes discovered in early plant growth stages can impact phenotypes such as terminal plant height and flowering times (Fig. 9B, C). This result is a proof of concept that UAVs can provide time-specific temporal phenotype data when plants are exposed to abiotic stress factors during growth (e.g. heat stress during flowering times in this study). Thus, relationships between plant abiotic stresses and candidate genes related to these stresses can be studied in greater biological detail with temporal data.

NDRBI was chosen as a model VI to further evaluate, due to having the highest ability of any RGB index to detect SNPs in the GWAS (Fig. 8A) as well as the highest heritability, lowest error, and highest R^2 (Fig. 3). Temporal genomic prediction was applied to a temporal value of NDRBI derived from multiple UAV flights during growth to examine whether the temporal phenotype is governed by genome-wide markers similarly or differently during growth and between environments. The results showed that there was a large variation in predicting temporal values of the NDRBI index across time points as well as trials in maize (Fig. 11A). The main reason is likely to be that the temporal values of the VI are influenced by both genetic and environmental factors as well as their interactions. This means that the relationship between the genomic data and the phenotype of interest may not be constant across different time periods due to changes in environmental conditions. Because the flight component explained the highest portion of explained variation in both trials, that provides some evidence of the dynamic relationship between temporal values of the VI and the environment (Fig. 11B). It was discovered that temporal NDRBI values belonging to time points between 75 and 99 DAPs were genomically predicted to be greater than those at 27–71 DAPs in both trials (Fig. 11). Among the most likely reasons is that the overall range of measurable phenotypic variation is larger in adult plants than in seedlings, as well as the range of heritable variation attributable to the genetic component. The resulting larger signal to measurement noise would be expected to result in predictions performing better in more mature plants. Another component might be that as the plant matures, the many genetic, environmental, and interaction factors that influence the phenotype of interest accumulate, making accumulated differences between genotypes more stable and predictable. In addition to these factors, the genetic architecture of the trait of interest may also change as the plant matures. For example, it is recognized that expression of certain genes may be turned on or off at different growth stages, thus the relative importance of different genetic markers for these genes may shift as the trait is expressed differentially over time (Zhou *et al.*, 2019). Overall, these findings demonstrated the variability in the prediction of temporal phenotype data under diverse environments since genomic markers had dynamic phenotypic effect sizes across different growth stages and environments (Fig. 12). Repetitive UAV surveys can be used to evaluate temporal prediction abilities of genome-wide markers for specific time points and environments. This approach advises using temporal values of agronomically important VIs in genomic prediction to better understand the predictability of genome-wide markers across times points and environments beyond using only one-time terminal phenotypic values (DTA, DTS, and PHT) as dependent variables.

Conclusion

FHTP incorporates novel technologies and applications such as UAVs and imaging sensors that can screen segregating populations with high precision, multiple measures, and a high time dimension. These high dimensional data allowed prediction and dissection of novel interactions between environment, plant, and underlying segregating genomic mechanisms. First, temporal phenomic spectral data belonging to multiple time points throughout growth predicted complex phenological traits (e.g. flowering time and plant height) across different environments, as well as helping to understand the stability of flowering. Second, temporal phenotype data were found to better dissect time-dependent associations between plant and abiotic stress. Finally, temporal reflectance bands led to the discovery of many loci, including known genes and a heat-related candidate gene only when heat stress reached a peak around flowering. Overall, temporal phenomic prediction and dissecting time-dependent association between temporal phenotype, genomic, and environmental data were demonstrated. Integration of high-throughput phenomic data into plant breeding and genomics will allow biological understanding to select novel complex traits and dissect the plant–environment interaction across plant development.

Supplementary data

The following supplementary data are available at *JXB* online.

Table S1. Vegetation indices used in this study with formulas and references.

Table S2. Discovered loci by GWAS with their SNP name, chromosome, genomic position, *P*-value, –log(*P*-value), minor allele frequency, effect size, flight time as days after planting time, vegetation index, phenotyping platform, trial, and candidate gene information.

Fig. S1. An orthomosaic belonging to a mid-growth stage of the irrigated trial; experimental design was illustrated via the orthomosaic.

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Fig. S2. One replication of 5-fold with 25 replications is illustrated for four prediction scenarios (CV1, CV2, CV0, and CV00).

Fig. S3. Linkage disequilibrium (LD) blocks of each chromosome were illustrated based on R^2.

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Author contributions

AA: conceptualization, data curation, formal analysis, investigation, methodology, original draft, review and editing (lead), supervision, validation, visualization; MK: investigation, methodology, and review and editing (supporting); SLA: investigation, validation, methodology, and review and editing (supporting); S.C.M: conceptualization, funding acquisition, methodology, project administration, supervision, resources, review and editing (supporting); DJ: data curation, methodology, validation, and visualization (supporting); RKWW: conceptualization, methodology, and review and editing (supporting); MK: conceptualization, methodology, and review and editing (supporting).

Conflict of interest

The authors have no conflicts to declare.

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Data availability

All data related to this manuscript are available in the Dryad Digital Repository https://doi.org/10.5061/dryad.1vhhmgqwm; (Adak *et al*., 2023c).

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